

Pleistocene climate oscillations and topographic barriers jointly shape stepwise phylogeographic divergence and polygenic adaptation to isothermality in *Bupleurum chinense*[☆]

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ABSTRACT

Understanding how widespread East Asian temperate taxa responded to Quaternary climatic oscillations and complex topography is central to explaining the assembly of regional biodiversity. Using the medicinal herb *Bupleurum chinense* as a model, we integrated phylogenomics, demographic inference, and genotype–environment association analyses to reconstruct its phylogeographic history and adaptive patterns. We identified four well-supported nuclear genetic lineages arranged in a stepwise geographic pattern from the southwest/northwest through central China to the northeast, indicating dispersal constrained by complex topographic barriers. Population structure was associated with both geographic distance and environmental differentiation, although the association with geographic distance was stronger. Demographic analyses support stepwise Pleistocene divergence from southwestern toward northern and northeastern China. This process was accompanied by asymmetric gene flow among adjacent lineages and episodes of range expansion followed by stabilization, suggesting climatically driven colonization and subsequent topographic isolation with intermittent secondary contact. Genotype–environment association analyses identified isothermality (Bio3) as the strongest environmental factor associated with genomic variation, and candidate loci were enriched in functions related to metabolic regulation and cellular stress response, consistent with polygenic adaptation to climatic instability. Chloroplast haplotypes showed high within-region diversity but limited sharing among regions, further supporting long-term lineage isolation and restricted interregional gene flow. Plant height also covaried spatially with Bio3, although this pattern is more consistent with environmentally responsive variation superimposed on partial genetic structuring than with simple fixed genetic divergence alone. Together, these results show that the present distribution and genomic architecture of *Bupleurum chinense* were jointly shaped by topographic barriers and climate-associated adaptation, providing new genomic insight into how geological heterogeneity and Pleistocene climate change influenced the evolution of widespread temperate taxa in East Asia.

1. Introduction

East Asia is globally recognized as a paramount center for plant biodiversity and endemism, offering a sophisticated natural laboratory

for studying the complex interplay between geological processes and biological evolution (Cai et al., 2023; Dupont-Nivet et al., 2025; Mi et al., 2021). Since the Neogene, and particularly throughout the Quaternary, this region has been characterized by dramatic environmental

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shifts driven by the cyclical nature of glacial-interglacial periods (Shukla et al., 2019). These paleoclimatic oscillations did not merely dictate the spatial migration of flora via long-distance dispersal and land-bridge formations; more importantly, they acted as a “species pump,” triggering intense lineage diversification within isolated micro-refugia through periodic environmental heterogeneity (Graham, 2018; Yu et al., 2019). Recent syntheses in the South China Karst (Tan et al., 2025) and Neotropical cycad lineages (Liu et al., 2025) have demonstrated that the synergy between orogenic uplift and climatic instability is a fundamental driver of modern biodiversity patterns. However, for widespread temperate species across East Asia, the genomic imprints of these spatiotemporal trajectories remain insufficiently explored, leaving a gap in our understanding of how broad-range taxa maintain genetic integrity or undergo divergence amidst shifting climates (Meng et al., 2025). More fundamentally, this raises a broader evolutionary question: how widespread species maintain genetic integrity while achieving local adaptation across complex topographic and climatic gradients.

Among the environmental factors shaping such evolutionary processes, climatic variability is increasingly recognized as a key driver of adaptive divergence. A central challenge in contemporary biogeography is identifying which climatic variables transcend simple correlation and act as genuine evolutionary drivers (Erwin, 2009). While traditional studies often emphasize mean annual temperature or total precipitation, growing evidence suggests that climatic stability and the frequency of extreme events can exert stronger selective pressures (Grant et al., 2017; Ummenhofer and Meehl, 2017). Isothermality (Bio3) and diurnal temperature range, both indicators of thermal stability, directly influence plant physiological rhythms, energy metabolism, and resource allocation strategies. In temperate herbs, adaptation to such thermal variability often manifests through suites of functional traits, including variation in plant height, leaf cuticle thickness, and phenological timing, which together buffer reproductive cycles against unpredictable temperature fluctuations (Liu et al., 2025). Nevertheless, empirical evidence remains limited regarding how these climate-associated traits are partitioned between genetic fixation and phenotypic plasticity, and how they contribute to population divergence in widespread temperate species.

Bupleurum chinense (Apiaceae), a widespread temperate herb prominently distributed across East Asia, provides an ideal system for examining these broader evolutionary processes (Downie et al., 2010; Huang et al., 2021; Song et al., 2025). Its distribution is remarkably extensive, spanning an immense ecological gradient from the rugged southwestern highlands, through the arid-to-semi-arid core of the Loess Plateau, to the fertile plains of Northeast China (Bai et al., 2022; Zhang et al., 2025). Such a vast latitudinal and longitudinal range implies that *B. chinense* may have navigated complex demographic bottlenecks and expansion events throughout its evolutionary history (Forester et al., 2016; Givnish, 2010; Overcast et al., 2023; Scheiner, 2013). Rather than serving solely as a single-species case study, *B. chinense* offers an opportunity to examine a broader evolutionary principle: how widespread species maintain genomic cohesion while evolving geographically structured adaptive responses across environmentally and topographically heterogeneous regions. Its broad distribution across highlands, plateaus, and plains also suggests prolonged exposure to contrasting thermal and hydrothermal regimes, which may have jointly shaped its demographic history and adaptive differentiation. Resolving the genomic architecture of *B. chinense* is therefore important not only for its conservation, but also for understanding how widespread temperate species persist, diverge, and adapt across heterogeneous East Asian landscapes (Becklin et al., 2016; Davis et al., 2005; Maguire et al., 2015). For *B. chinense*, the transition from high-amplitude temperature fluctuations in highland habitats to the relatively buffered thermal regimes of lowland plains may have imposed strong selective pressures across its genome. Such environmental contrasts likely required trade-offs between vegetative growth and stress resilience; for example, populations in highly variable environments may favor reduced stature and

reinforced cell-wall biosynthesis to withstand sudden frost or heat stress (Climent et al., 2024). Adaptation to temperature variability may therefore have been an important prerequisite for the successful colonization of the North China Plain and adjacent lowland regions. However, empirical data remain scarce regarding the extent to which these adaptive traits are genetically fixed versus environmentally plastic, and how they contribute to population divergence in widespread temperate herbs.

In summary, *B. chinense* is not only a vital ecological component but also a pivotal model for deciphering the evolutionary history of East Asian temperate flora. By integrating high-resolution population genomics across its entire distribution with genotype–environment association (GEA), this study aims to address three specific questions. First, did the major lineages of *B. chinense* diverge stepwise during the Quaternary, and was this process associated with climatic oscillations? Second, does isothermality (Bio3) represent a key selective factor shaping genomic variation and phenotypic trade-offs, especially those related to plant height, through polygenic adaptation? Third, to what extent does variation in plant height reflect environmentally responsive plasticity versus genetic differentiation, and how is this pattern associated with Bio3-related adaptation in *B. chinense*? We hypothesize that adaptation to thermal stability has been central to the persistence and divergence of *B. chinense* across broad environmental gradients. To test these ideas, we combined transcriptome-derived single-copy orthologs, WGS-based demographic inference, and dual-algorithm validation of GEA signals to link historical divergence, climate-associated adaptation, and present-day population structure within a unified framework. Our findings provide a genomic perspective on how widespread temperate species persist, diverge, and adapt across the heterogeneous landscapes of East Asia, while also offering a basis for assessing their vulnerability under future climatic change.

2. Materials and methods

2.1. Material collection and sequencing

We sampled a total of 24 populations throughout the natural distribution ranges of *Bupleurum chinense* including 81 individuals of *B. chinense* (3–5 individuals per population) and 26 additional *Bupleurum* species used for comparative analyses. Fresh leaves were dried in silica gel in the field and stored in -20°C for DNA extraction. Voucher specimens were deposited in the Herbarium of Sichuan University (SZ) (Table S1). The genomic DNA samples were first fragmented, followed by end polishing, A-tailing, ligation with full-length sequencing adapters, and polymerase chain reaction (PCR) amplification. After quality assessment and quantification, the qualified libraries were sequenced by the sequencing provider using Illumina NovaSeq PE150 platforms. On average, $\sim 28\times$ sequencing depth per individual was obtained. Mapping and coverage statistics were summarized using SAMtools, with most samples showing high mapping rates ($>95\%$) and genome coverage exceeding 90%. In addition, fresh leaves from *Chamaesium paradoxum* and *B. chinense* were collected for RNA-seq. For each species, RNA-seq was generated from a single individual. Library preparations were conducted at Shanghai Personalbio Technology Co., Ltd. using the Illumina NovaSeq PE150 platform. RNA samples were processed by Shanghai Personalbio Technology Co., Ltd., where RNA extraction, library construction, and sequencing were conducted. The libraries were prepared using an NEBNext Ultra RNA Library Prep Kit for Illumina according to the manufacturer's recommendations. Sequencing was performed by Shanghai Personalbio Technology Co., Ltd. on the Illumina NovaSeq platform with a PE150 read configuration. After quality filtering, high-quality clean reads were obtained for downstream analyses, with Q30 values exceeding 90%.

2.2. Transcriptome assembly, ortholog inference and phylogenetic analyses

Filtered RNA-seq reads from *C. paradoxum* and *B. chinense* were assembled into contigs de novo using the Trinity v2.15.1 tool (Grabherr et al., 2011; Haas et al., 2013) with default parameters (`--seqType fq --kmer-size 25 --trimmomatic`). Protein-coding regions were subsequently identified using the TransDecoder v5.7.1 (Haas et al., 2013). Ortholog inference was conducted using Orthofinder v2.5.4 (Emms and Kelly, 2015) with the parameters “-M msa -S diamond -A mafft” to identify putative orthologs. From the resulting orthologous groups, 3446 sequences of single-copy orthologs (SCOs) sequences were selected as references for downstream analyses.

Using the above SCO sequences as references, we assembled the *Bupleurum* SCO genes datasets from whole-genome resequencing data using GeneMiner2 (Xie et al., 2024; Zhang et al., 2022b). Each assembled gene was aligned separately using MAFFT v7.52 (Katoh and Standley, 2013), and the aligned sequences were trimmed using TrimAl v1.2 (Capella-Gutiérrez et al., 2009). Ultimately, we selected 2103 SCO genes (Table S2) for phylogenetic analysis. We employed the coalescent method to construct phylogenetic trees. The maximum likelihood (ML) analyses of the SCO genes were conducted using FastTree 2.1 (Price et al., 2010) with 10,000 bootstrap replicates under the GTR + G model. Single-gene tree data sets generated from these analyses were subsequently used to construct a species tree with ASTRAL-III (Zhang et al., 2018).

2.3. SNP calling of *Bupleurum chinense*

High-quality raw reads were mapped to the published reference genome of *Bupleurum chinense* (Zhang et al., 2022a). Using the Burrows-Wheeler Alignment-Maximal Exact Match module with default parameters in bwa-mem2 v2.21 (Vasimuddin et al., 2019). The mapping results were converted to BAM and processed with SAMtools v1.8.4 (Danecek et al., 2021) to generate analysis-ready files coordinate sorting, name sorting, mate-information fixing, re-sorting to coordinate order, duplicate marking, and BAM indexing and read-mapping and coverage statistics were likewise summarized with SAMtools v1.8.4 (Danecek et al., 2021). SNP calling was performed with bcftools by piping mpileup (`-q 30 -Q 30 -d 1000`) into call -mv, yielding bgzip-compressed, indexed VCFs. To minimize potential false positives and obtain high-quality genotype calls, variants were hard-filtered with bcftools view using `QUAL <30 || DP < 10 || DP > 100`. Finally, we obtained Dataset 1.

Using Plink v1.90b6.21 (Purcell et al., 2007), Dataset 1 was filtered by removing SNPs with missing rate > 0.20 and minor allele frequency (MAF) < 0.01. The same SNP filtering thresholds were applied uniformly across all populations, regardless of habitat type or elevation, to reduce potential biases arising from variable data quality among high-altitude and plain populations. Retained variants were converted to Plink binary format (.bed/.bim/.fam). LD pruning with a 50-SNP window, 10-SNP step, and $r^2 = 0.2$ (`--indep-pairwise 50 10 0.2`) was then applied, and the pruned loci were extracted to generate Dataset 2. Using Dataset 1, SNPs identified as candidate environment-associated loci in the GEA analyses were removed. SNPs located within coding regions and within 2000 bp upstream and downstream of annotated genes were further excluded based on the reference genome annotation. The remaining loci were filtered to retain SNPs without missing data across all individuals, and the retained loci were extracted to generate Dataset 3.

2.4. Population structure and demographic history

Dataset 2 was used for population structure analyses. PCA was conducted in Plink v1.90b6.21 (Purcell et al., 2007) with default settings, and results were visualized in R with ggplot2 (Hadley, 2016). To

infer the optimal number of ancestries (K), each K from 2 to 10 was tested in Admixture v1.3.0 (Alexander et al., 2009), and the best K value was identified based on the cross-validation error. In addition, a neighbor-joining phylogeny with bootstrap support was constructed using VCF2Dis v1.50 (<https://github.com/BGI-shenzhen/VCF2Dis>) under default parameters.

Fastsimcoal2 v2.7 (Excoffier et al., 2021) was used to explore the evolutionary history of *B. chinense* based on Dataset 3. To ensure reliable estimation of the site frequency spectrum (SFS), only loci without missing data across all individuals were retained for fastsimcoal2 analyses. We specified 14 alternative divergence scenarios (Fig. S2) to identify the best-fitting model, which mainly differed in the order of lineage divergence, the presence or absence of gene flow among lineages, and changes in effective population size through time. Remaining simulation settings and procedures followed previous studies (Chen et al., 2021; Excoffier et al., 2013). Each model was run 50 times, with 100,000 simulations per run to compute the composite likelihood, followed by 40 expectation conditional maximization (ECM) cycles. The model with the minimum Akaike information criterion (AIC) was chosen as optimal; Model 10 was ultimately selected. Parameter point estimates were taken from the run with the highest maximum composite likelihood, and 100 parametric bootstrap replicates were performed by simulating SFS under these point estimates to obtain confidence intervals (CIs).

Two individuals from each of the four groups of *B. chinense* (the four genetic groups inferred from the population structure analyses described above) were used to infer the dynamics of historical effective population size by the PSMC model (Li and Durbin, 2011). Merged VCFs were processed with bcftools v1.10.2 (Danecek et al., 2021) using view -S for sample extraction, norm -d all -f to remove overlapping variants, and filter -IndelGap to exclude SNPs proximal to indels, yielding high-quality variant sets for consensus sequence construction (Danecek et al., 2021). PSMC was then run with parameters `-N25 -t15 -r5 -p “4 + 25*2 + 4 + 6”` to obtain demographic estimates. Generation time was set to 5 years based on both published studies (Song et al., 2025) and field observations, and the mutation rate μ was set to 4.57×10^{-8} per site per generation, estimated according to genome-wide Ks of collinear orthologous gene pairs between *B. chinense* and *B. scorzoniferifolium*, which are dated to have diverged at ~5.14 Mya (Song et al., 2025).

In addition, population genetic statistics were calculated using VCFtools (Danecek et al., 2011), including nucleotide diversity (π), Tajima's D, and population divergence (F_{ST}), using a sliding-window size of 200 kb.

2.5. Environmental data acquisition and analyses

We assembled four categories of environmental data (climate, soil, elevation, and human impact) comprising 50 variables in total. Climate and elevation were obtained from WorldClim (<https://www.worldclim.org/>), the ENVIREM environmental raster database (<http://envirem.github.io/>), and CGIAR-CSI (<https://cgiarcsi.community/>). Soil variables were sourced from the FAO World Soil Database v2.0 (<https://www.fao.org/>) at 1-km resolution. Human Influence Index and Human Footprint data were downloaded from CIESIN, Columbia University (<http://www.ciesin.columbia.edu>). Environmental values were extracted directly from raster layers at each sampling-point coordinate. WorldClim variables were obtained at 30 arc-seconds resolution, whereas ENVIREM and soil variables were obtained at 1-km resolution. Detailed information for all 50 environmental variables is provided in Table S3. PCA was conducted using the R package factoextra (Kassambara and Mundt, 2017).

Partial Mantel tests were used to assess correlations among three matrices: genetic divergence, geographic distance, and environmental distance (Diniz-Filho et al., 2013). An individual-wise identity-by-state (IBS) similarity matrix was generated in Plink and converted to a genetic

distance matrix for Mantel testing. Latitude/longitude coordinates of sampling sites were converted into a geographic distance matrix among individuals. Environmental variables were z-score standardized and a Pearson correlation matrix was computed; highly correlated variables ($|r| > 0.8$) were removed. Based on the filtered variables, Euclidean environmental distance matrix among individuals was calculated. Finally, partial Mantel tests were implemented with a Python workflow to evaluate associations among individual-level genetic, geographic, and environmental distance matrices, with significance assessed by 999 permutations.

2.6. Genotype-environment association analysis

To reduce multicollinearity among environmental predictors prior to GEA, we performed Spearman's rank correlation on the 50 environmental variables described above. For each pair with $|r| > 0.8$, one variable was retained, yielding a final set of 20 variables (Table S4).

For these 20 variables, we combined two different approaches to identify environment-associated SNPs across the genome based on Dataset 2. First, we used BayPass v2.3.1 (Gautier, 2015) to perform GEA. Under the core model, all filtered SNPs were used to estimate population structure and obtain the covariance matrix. This covariance matrix was then supplied to the auxiliary covariate model to explicitly account for shared demographic history and background population structure while testing SNP-environment associations, thereby reducing confounding caused by geographic and ecological contrasts among populations, including those between high-altitude and plain regions. Bayes factors (BF_{mc}) were computed for each model, and environmental covariates were scaled with `-scalecov` option. To define significance thresholds, we simulated pseudo-observed data (POD) comprising 1000 SNPs with `simulate.baypass` using the core-model covariance matrix, and analyzed each environmental variable as above; the 99th percentile of POD BF values was used as the significance threshold.

Second, we applied a univariate latent factor mixed model implemented in LFMM_CL v1.5 (Frichot and François, 2015) to test associations between allele frequencies and the 20 environmental variables. Guided by the three ancestry components inferred with Admixture v1.3.0 (Alexander et al., 2009), we set the number of latent factors to 3. These latent factors were used to capture unobserved population structure and thus to minimize the risk that environmental contrasts, including those associated with elevation, would be conflated with background genetic structure. For each environmental variable, five independent MCMC runs were conducted with 5000 burn-in iterations followed by 10,000 sampling iterations. For each SNP, *p*-values were combined across the five runs and adjusted using a 1% false discovery rate (FDR) to identify significant associations.

We then intersected candidate SNPs detected by BayPass and LFMM to retain only associations supported by both methods, thereby reducing method-specific false positives and increasing robustness when comparing populations from environmentally contrasting regions. The intersected candidate set was visualized, and GO enrichment was performed using the R packages CMplot (Yin et al., 2021) and clusterProfiler (Wu et al., 2021), with significance defined as $q \leq 0.05$ (FDR-adjusted *P*-value). Finally, the 6078 candidate SNPs were annotated using SnpEff v4.383 (Cingolani et al., 2012), setting `-ud 2000` to define 2000-bp upstream and downstream windows around genes while keeping other parameters at default.

2.7. Haplotype network analysis

We assembled the chloroplast genome of *B. chinense* using GetOrganelle v1.7.7.0 (Jin et al., 2020) with default parameters. Plant height was recorded for each population. Bio3 (isothermality), a WorldClim bioclimatic variable representing the ratio of the mean diurnal temperature range to the annual temperature range ($\times 100$), for each population was extracted in R, and K-means clustering (Likas et al., 2003)

was performed using Bio3, longitude, and latitude as input variables. Prior to clustering, all variables were z-score standardized. The number of clusters was set to $k = 2$, partitioning samples into two major groups (A/B) (Table S5).

Genes associated with plant height among the Bio3 significant candidates were then used as references. Using GeneMiner2 (Xie et al., 2024; Zhang et al., 2022b), we assembled the corresponding “plant-height genes” for each *B. chinense* sample from raw resequencing reads. The assembled sequences were aligned to the reference, followed by multiple sequence alignment; alignments were trimmed with trimAl (Capella-Gutiérrez et al., 2009). The resulting data were imported into NetST (<https://github.com/sculab/NetST>) for haplotype analyses, and both plant height and Bio3 group assignments were annotated on the haplotype network.

3. Results

3.1. Phylogenetic relationship and population structure

A total of 26 *Bupleurum* species and 81 individuals of *B. chinense* were included in the phylogenetic analyses. The nuclear dataset supported *B. chinense* as a monophyletic lineage that is sister to *B. malconense*, *B. scorzoniferifolium*, and *B. falcatum*, located at the top of the phylogenetic tree (Fig. S3).

Genetic relationships among *B. chinense* populations were reconstructed using a maximum-likelihood phylogeny, principal component analysis (PCA), and admixture (Fig. 1). The PCA explains 12.55% and 10.06% of the variance along PC1 and PC2, respectively. Along PC1, the Group 1 and Group 3 are separated from the remaining samples. Admixture revealed clear within species structure; the cross-validation error was minimized at $K = 4$. At this resolution, four ancestry components corresponded to geographic groupings: Group 3 showed largely unmixed ancestry and were concentrated in central China. In contrast, Group 1, which is distributed in southwestern China, and Groups 2 and 4, which show higher proportions of admixture, are mainly distributed in central and northeastern China. The populations of YZ, WSY, WH, and CHS, distributed in the transitional zone, are located at the base of the G2 branch in the phylogenetic tree, exhibiting a higher degree of genetic admixture, which may reflect the complex genetic background and gene flow patterns of Group 2 (Fig. 1).

3.2. Population demographic history and genetic diversity

Among the 14 demographic scenarios tested (Fig. S2), the model with the lowest AIC (Model 10) was selected as optimal (Table S6). This scenario recovered a stepwise split: the southwestern montane group (Group 1) diverged first at $T1 \approx 355.7$ kya, followed by the west-central transition group (Group 2) at $T2 \approx 124.1$ kya, and finally a split between central China (Group 3) and North/Northeast China (Group 4) at $T3 \approx 34.22$ kya (Fig. 2E), mirroring spatial turnover. Contemporary effective population was larger in Group 2 and Group 4 and smaller in Group 1 and Group 3 (Fig. 2). The best-fit model also supported asymmetric post-divergence gene flow that is widespread and directionally biased among adjacent groups, with the strongest rates between neighboring pairs (Group 1–Group 2, Group 2–Group 3, Group 3–Group 4).

We then applied PSMC modelling to reconstruct the population-size changes history of *B. chinense*. All groups maintained low effective population size in the early phase (>1 Ma), followed by expansions that peaked at ~ 300 – 500 kya and then declined toward recent stabilization (Fig. 2C). Notably, Group 1 retained the lowest effective population size, with limited expansion and slow recovery, suggesting stronger bottlenecks or prolonged small-population effects. Group 4 underwent the most pronounced expansion, peaking ~ 300 – 500 kya, consistent with access to favorable niches during this interval. Groups 2 and 3 showed intermediate, broadly similar histories with mid-Pleistocene fluctuations, in line with their low *F_{ST}* and higher connectivity.

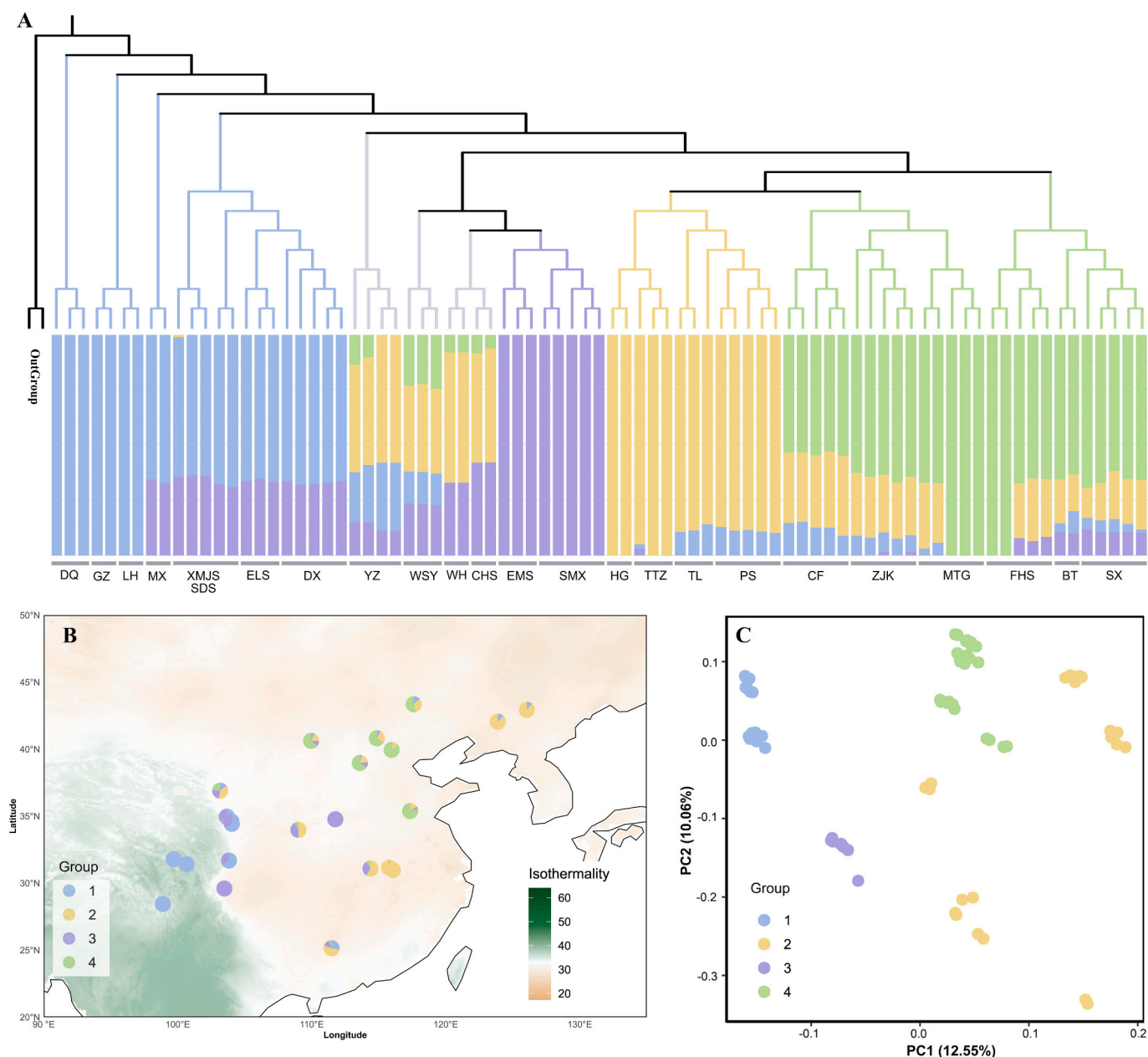


Fig. 1. Population genomic analyses of *B. chinense*. A). Neighbor-joining phylogenetic tree based on SNPs, using *B. franchetii*, *B. gracillimum*, *B. rotundifolium* as outgroups. B). ADMIXTURE analysis with the optimal K value ($K = 4$), which minimizes cross-validation error. Each vertical bar represents an individual sample, with the x-axis denoting sample localities. Colors within the bars indicate putative ancestral backgrounds, while the y-axis quantifies the proportion of ancestry. C). Geographic distribution of the analyzed samples, with colour codes representing their various localities. D). Principal component analysis (PCA) showing the first two principal components. Colors represent phylogenetic groupings.

Within *B. chinense*, Group 3 exhibited the highest nucleotide diversity (π), whereas Group 1 showed the lowest median π (Fig. 2). Tajima's D was highest in Group 4, followed by Group 1. Pairwise differentiation further supported these patterns: the greatest divergence occurred between Group 1 and Group 4 ($F_{ST} = 0.134$), consistent with their occurrence in distinct regions where environmental contrasts and geographic barriers restrict gene flow; by contrast, Group 2 vs. Group 3 showed the lowest differentiation ($F_{ST} = 0.078$ reflecting weak genetic divergence between adjacent ranges under similar ecological regimes. These results accord with Admixture and PCA (Fig. 1).

3.3. Analysis of environmental data

Partial Mantel tests showed significant associations of both

geographic distance (IBD) and environmental distance (IBE) with genetic divergence in *B. chinense*. Controlling for environmental distance, genetic distance remained strongly correlated with geographic distance ($r = 0.501$, $P = 0.001$; Fig. 3A). In contrast, when geographic distance was controlled for, genetic distance was still positively correlated with environmental distance, although the correlation was weaker ($r = 0.224$, $P = 0.001$; Fig. 3B). Together, these results suggest that both geographic and environmental distances are associated with genetic divergence, although the geographic association was stronger under the partial Mantel framework.

In the PCA based on 50 environmental variables, the first three components explained 68.19% of the total variance (PC1, 35.56%; PC2, 18.83%; PC3, 13.80%; Fig. 3C, D). In the PC1–PC2 space, Group 1 was mainly distributed at positive PC1 values and lower PC2 values, whereas

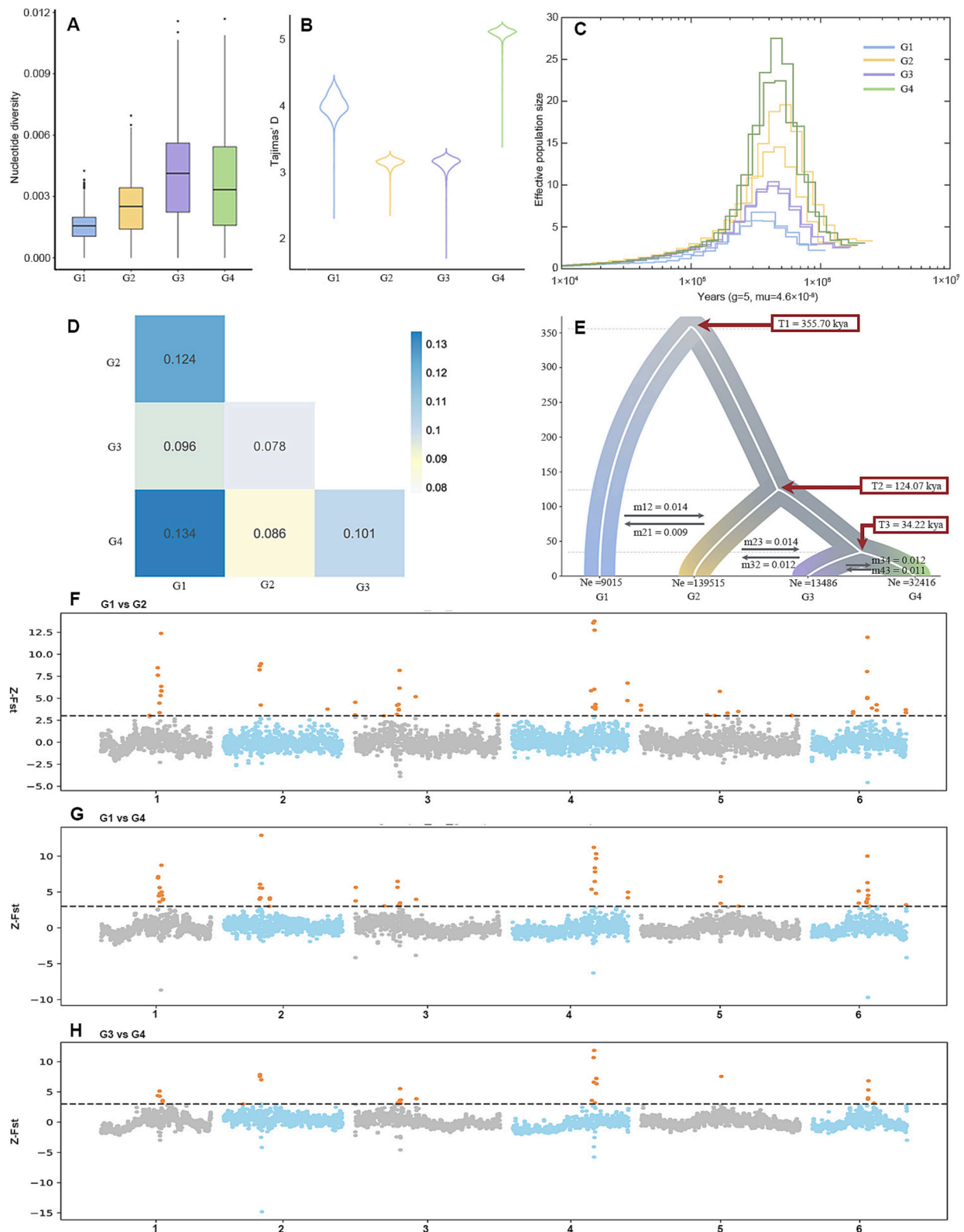


Fig. 2. Demographic history of different genetic groups of *B. chinense*. A). Boxplot of nucleotide diversity. B). Violin plot of Tajima's D. C). Pairwise sequentially Markovian coalescent (PSMC) estimates of changes in effective population size (N_e) through time. A generation time of 5 years and a mutation rate μ of 4.57×10^{-8} per site per generation were assumed. D). Heatmap of pairwise F_{ST} estimates between groups. E). Graphical summary of the optimal demographic model determined by fastsimcoal2. Estimates of gene flow between lineages are presented as the migration rate per generation. F). Z values of F_{ST} across the genome in the comparison between Group 1 and Group 2. G). Z values of F_{ST} across the genome in the comparison between Group 1 and Group 4. H). Z values of F_{ST} across the genome in the comparison between Group 3 and Group 4.

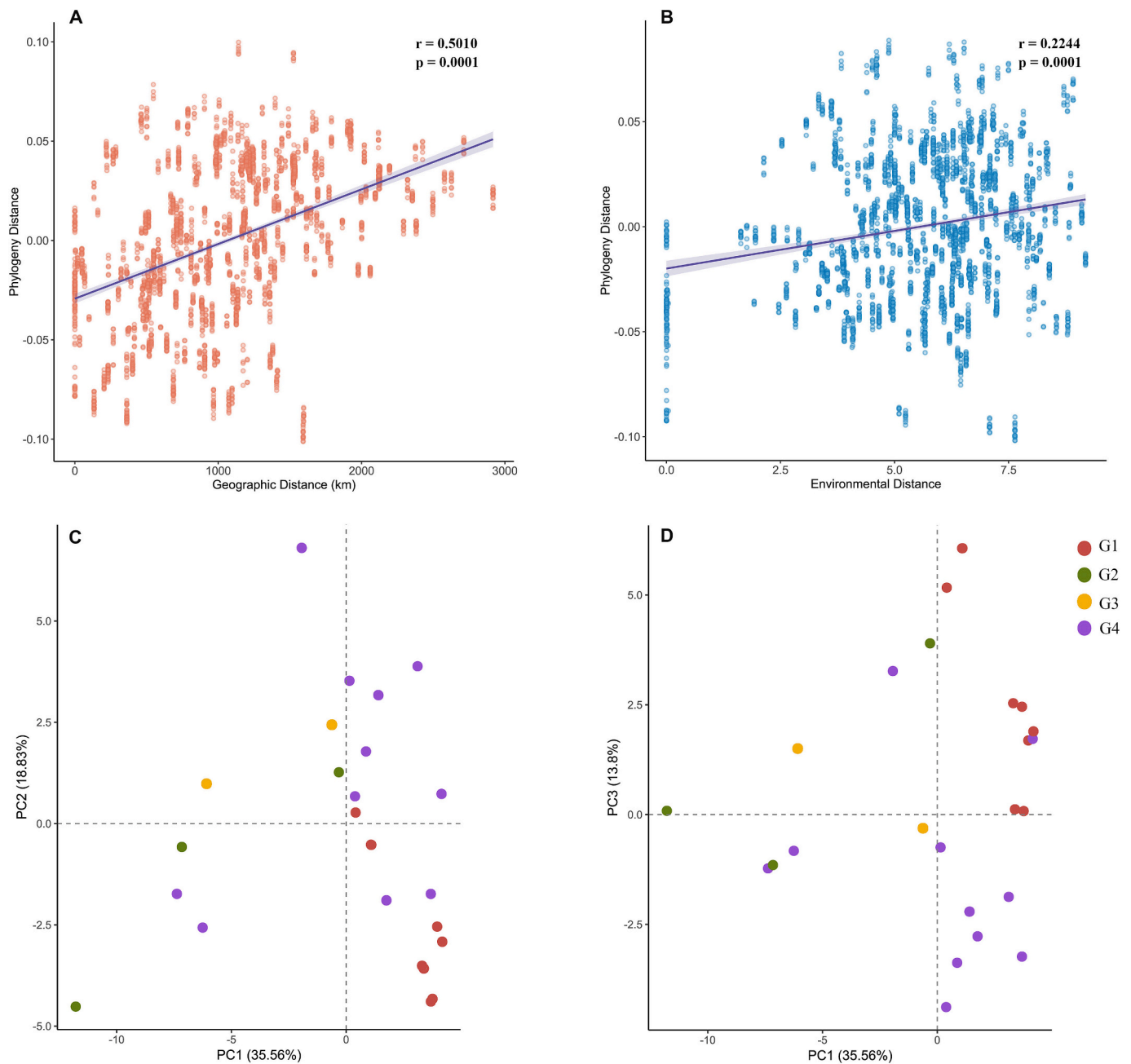


Fig. 3. Partial Mantel tests and environmental principal component analysis of *B. chinense*. A). Correlation between phylogenetic distance and geographical distance. B). Correlation between phylogenetic distance and Environmental distance. C). Principal component analysis (PCA) results for 50 environmental factors. PC1 and PC2 explained 35.56% and 18.83% of the total variation. D). Principal component analysis (PCA) results for 50 environmental factors. PC1 and PC3 explained 35.56% and 13.8% of the total variation.

Groups 2 and 3 were largely distributed at negative PC1 values; Group 4 occupied a broader range of PC1 values but was generally separated from Group 1 along PC2 (Fig. 3C). In the PC1–PC3 space, Group 1 and Group 4 were more clearly differentiated along PC3, whereas Groups 2 and 3 remained largely overlapping (Fig. 3D). Overall, these patterns indicate substantial environmental differentiation among groups, with the strongest contrast occurring along PC1. PC1 was strongly associated with precipitation-related variables, including precipitation of the driest month (Bio14), precipitation of the driest quarter (Bio17), precipitation of the coldest quarter (Bio19), annual precipitation (Bio12), and precipitation of the wettest quarter (Bio16), suggesting that precipitation gradients were a major contributor to environmental differentiation among groups (Table S7).

3.4. Genotype-environment association analysis

Using the latent factor mixed model (LFMM) and BayPass methods, we identified 54,647 and 51,350 SNPs, respectively, that were associated with at least one of the 20 environmental variables (Table S7). Of the 51,350 significant variants identified by BayPass, 6078 were also identified by LFMM. We retained 6078 core adaptive SNPs associated with at least one environmental factor (Fig. 4). Among these 6078 SNPs, 1132 (5.97%) were exonic and 3233 (17.06%) intronic, with 4764 (25.13%) located in intergenic regions; within exons, 627 (55.39%) were nonsynonymous and 505 (44.61%) were synonymous. Genes in regions harboring significant SNPs were additionally annotated with eggNOG-mapper, and GO enrichment revealed significant

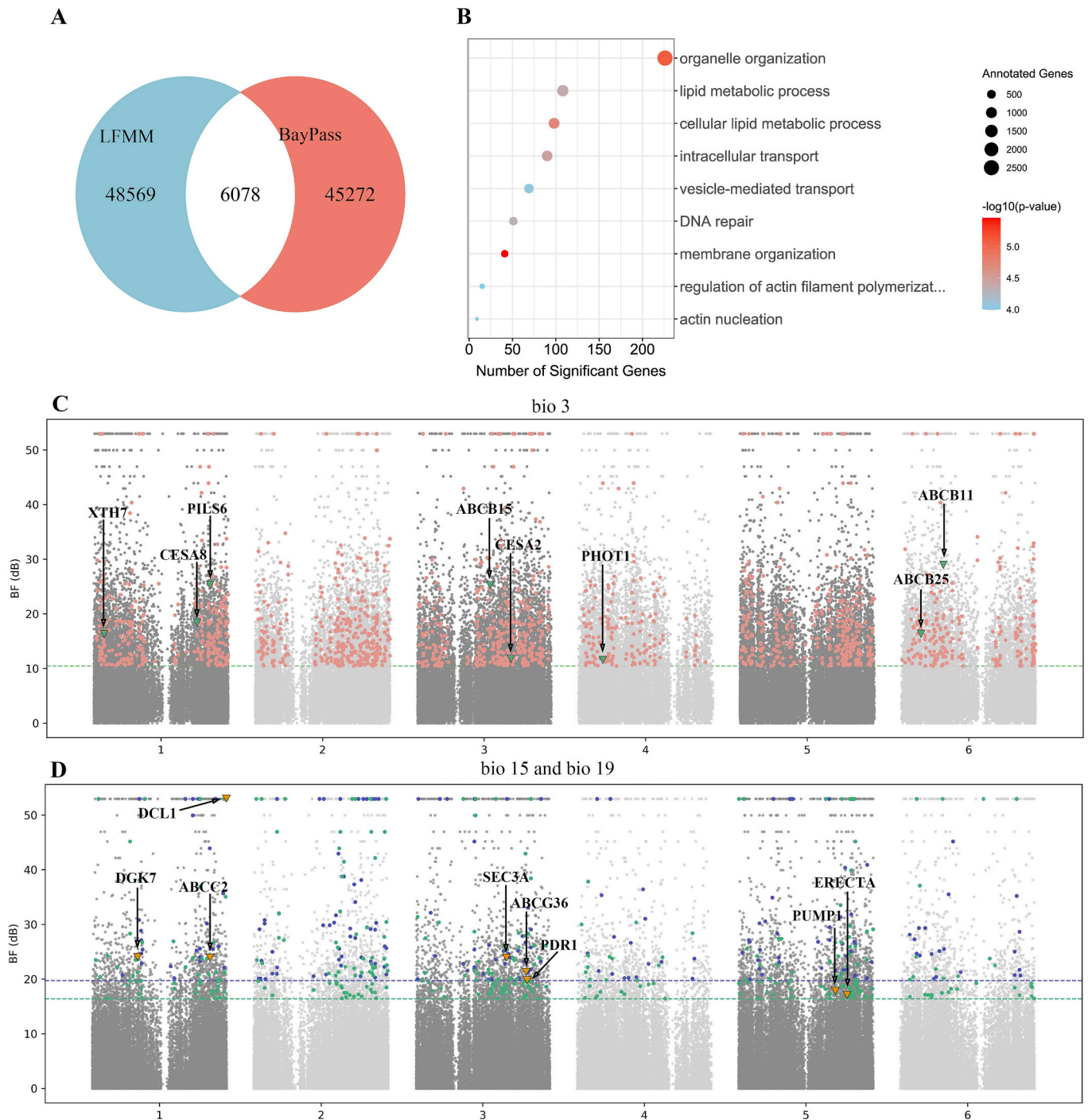


Fig. 4. Genotype-environment association analysis of *B. chinense*. A). The numbers of SNPs exhibiting strong associations with 20 environmental variables, as detected by BayPass and LFMM, and the number of SNPs shared between the two methods. B). Gene Ontology enrichment analysis of environment-associated significant SNPs. C). Genotype-environment association analysis for the environment factor Bio3. SNPs on different chromosomes are represented in alternate colors. The dashed line is the BayPass threshold line, and the red dots are significant SNPs. The green triangles represent SNPs detected as plant height. D). Genotype-environment association analysis for the environment factor Bio15 and Bio19. SNPs on different chromosomes are represented in alternate colors. The dashed line is the BayPass threshold line, and the blue and green dots are significant SNPs for Bio15 and Bio19, respectively. The orange triangles represent SNPs detected as transport and vesicle systems alongside kinase cascades and membrane homeostasis and defense and repair. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

overrepresentation of processes related to stress responses, lipid metabolism, organelle and membrane organization, intracellular localization and transport, DNA repair, and cytoskeleton organization and regulation (Fig. 4; Table S9).

Across variables, the number of significant SNPs varied widely from 481 for GHIL to 29,643 for Bio3. Overall, climate variables (e.g., Bio3,

Bio15, Bio19) yielded more associations than others, indicating that temperature and precipitation are major ecological drivers of genomic variation and adaptation, consistent with the species' broad distribution across contrasting climates. Notably, Bio3, which indexes the relative amplitudes of annual vs. diurnal temperature ranges, showed the strongest signal, suggesting extensive genomic responses to large

day–night thermal contrasts.

Projecting Bio3-significant SNPs onto gene bodies identified 533 genes; 99.2% received reliable Swiss-Prot annotations. The candidate genes are enriched in modules related to core metabolism,

transmembrane transport, transcriptional and chromatin regulation, lipid metabolism, stress and defense responses, and vesicle trafficking; representative genes include BTAF1, DGK7, ABCC2, BIG5, TPS1, and RRP44A (Table S10). These results indicate that adaptation to diurnal

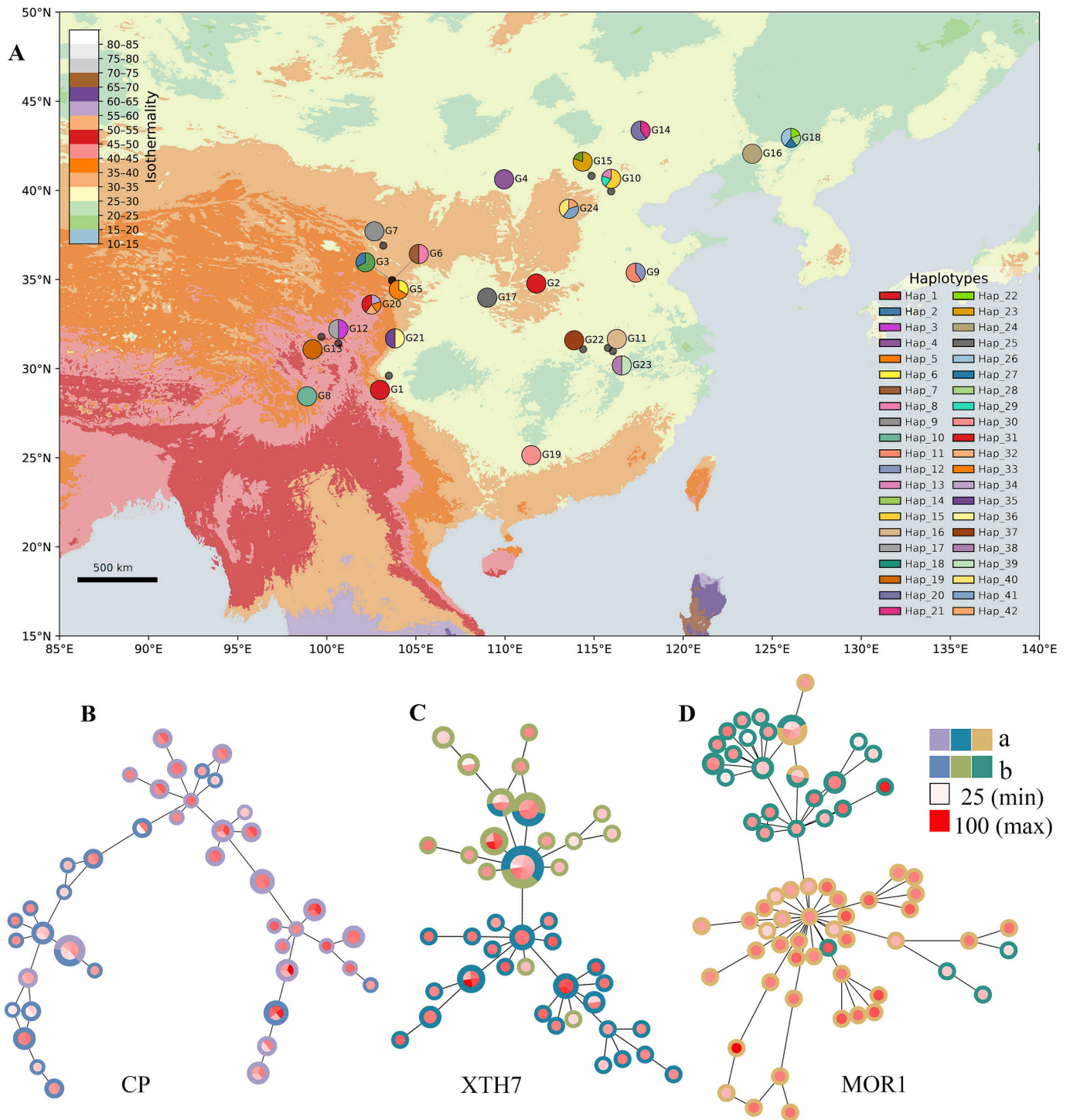


Fig. 5. Haplotype analysis of *B. chinense*. A). Geographical distributions of cpDNA haplotypes. B). Statistical parsimony network of chloroplast genome haplotypes. Circle sizes are roughly proportional to individual numbers of haplotypes. The inner circle indicates plant height, with darker shades representing taller plants. The outer circle represents the bio3 variable, blue indicates high bio3 values and purple indicates low bio3 values. C). Statistical parsimony network of XTH7 Gene. Circle sizes are roughly proportional to individual numbers of haplotypes. The inner circle indicates plant height, with darker shades representing taller plants. The outer circle represents the bio3 variable, green indicates high bio3 values and blue indicates low bio3 values. D). Statistical parsimony network of MOR1 Gene. Circle sizes are roughly proportional to individual numbers of haplotypes. The inner circle indicates plant height, with darker shades representing taller plants. The outer circle represents the bio3 variable, green indicates high bio3 values and blue indicates low bio3 values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

temperature range and seasonal thermal regimes is a multilevel, coordinated genetic process.

We further compared candidate gene sets for Bio3, Bio15, and Bio19. All three showed pronounced polygenicity and converged on core modules, including transmembrane transport, primary metabolism, transcriptional and chromatin regulation, vesicle trafficking, and stress and defense. Concurrently, the significantly associated genes displayed distinct, factor-specific functional signatures. For Bio3, genes coupling metabolism with epigenetic regulation (e.g., TPS1, DGK7, BTA1, EBS, MYC2) were overrepresented, indicating that diurnal and seasonal temperature regimes drive adaptation through coordinated adjustments in carbon and lipid metabolism together with chromatin remodeling. Bio15 emphasized transport and vesicle systems alongside kinase cascades (e.g., ABCC2, ABCG36, PDR1), consistent with physiological demands for rapid solute allocation and timely deployment of receptors and transporters under fluctuating water availability. Bio19 highlighted membrane homeostasis and defense and repair (e.g., ERECTA, PUMP1, DGK7) under combined cold and moisture stress. We further identified 30 candidate genes associated with plant height among the BIO3-significant loci (Table S11). These candidates are mainly involved in transcriptional and growth regulation, auxin (IAA) biosynthesis, transport, and signaling, sugar and energy metabolism, and cell-wall dynamics and cell elongation. Within the proximal modules that directly drive elongation, we detected five genes for auxin transport (ABCB15, PHOT1, ABCB11, PILS6, ABCB25) and three genes for cell-wall extension (CESA8, XTH7, CESA2) (Fig. 4).

3.5. Haplotype analysis

We assembled chloroplast genomes for 81 individuals of *B. chinense* from 24 populations (Fig. S4), with an average genome size of 152,704 bp. The haplotype network is shown in Fig. 5A, B. In total, 42 haplotypes were identified (Fig. 5; Table S12), encompassing 27,154 polymorphic sites. Haplotype distributions exhibit marked geographic structure. Populations in the southwest and adjacent regions (e.g., G3, G5, G6, G12, G20) generally harbor high haplotype diversity with multiple haplotypes co-occurring within populations, yet no haplotypes are shared among these southwestern populations. Eastern populations (e.g., G15, G24, G10, G18) likewise show high diversity, each composed of private haplotypes without inter-population sharing (Fig. 5). By contrast, central and some southern populations (e.g., G2, G19, G22) are dominated by one or a few haplotypes and display lower diversity. Overall, the southwest and the east emerge as diversity hotspots, whereas the general lack of haplotype sharing among regions indicates restricted gene flow and strong lineage isolation.

K-means clustering based on Bio3 and geographic coordinates partitioned populations into an eastern, low-Bio3 group (A) and a western, high-Bio3 group (B). Mean Bio3 values were 27.84 for group A and 34.06 for group B. K-means group identity and plant height were annotated on the haplotype network: the inner circle shading reflects plant height with darker shading indicating taller individuals, and the outer ring denotes k-means group assignment. Group A shows a mean height of 77.6 cm, with a maximum of 130 cm, whereas group B averages 53.3 cm with most individuals between 25 and 80 cm. The haplotype network is broadly divided into two clusters, and both the chloroplast data (Fig. 5B) and haplotypes of plant-height candidate genes XTH7 and MOR1 (Fig. 5C, D) consistently indicate taller plants in the eastern, low-Bio3 group (A) compared with the western, high-Bio3 group (B).

4. Discussions

4.1. Spatiotemporal evolution of *Bupleurum chinense*: pleistocene climate oscillations and “stepwise” divergence

Our population-genomic analyses support a scenario of stepwise phylogeographic divergence in *B. chinense* during the Pleistocene,

extending from Southwest China toward North China and Northeast China. SNP-based phylogenetic inference, ADMIXTURE, and PCA converge on four major genetic lineages (Group 1 - Group 4) (Fig. 1), each exhibiting clear geographic cohesion and sequential turnover. In the context of the sampling distribution (Fig. 1B), these lineages align broadly with China's major geomorphic tiers, from the southwestern mountains through the central transition zone to the Loess Plateau and areas farther east and north, suggesting that key mountain systems and plateau margins have acted as persistent dispersal-resistance surfaces and lineage boundaries. Partial Mantel tests further suggest that geographic distance contributed more strongly than environmental distance to the observed genetic structure, although both were significantly associated with genetic divergence (Gao et al., 2023; Han et al., 2017).

Demographic modelling with fastsimcoal2 places the major lineage splits largely within the Pleistocene, consistent with diversification under repeated glacial-interglacial oscillations. The best-fit model supports early separation of the southwestern lineage (Group 1; $T_1 \approx 355.7$ kya), followed by divergence of the transitional lineage (Group 2; $T_2 \approx 124.1$ kya), and the most recent split between central and northern/northeastern lineages (Group 3 vs. Group 4; $T_3 \approx 34.22$ kya). This temporal sequence, coupled with the spatial ordering of lineages, is most consistent with a history of alternating range expansion, establishment, and isolation/secondary contact rather than a smooth clinal pattern. The earliest divergence (T_1) corresponds to Marine Isotope Stage (MIS) 10 (~ 374 – 337 ka), a significant glacial interval. During this period, intensified cooling and habitat fragmentation likely forced *B. chinense* into isolated montane refugia in Southwest China, initiating the first step of lineage separation. In contrast, T_2 occurs during MIS 5e (~ 130 – 115 ka), the peak of the Last Interglacial. This warm interval likely facilitated northward and eastward range expansion from southwestern centers into the west-central transition zone. The subsequent shift toward stadial conditions may have trapped these colonizing populations, establishing G2 as a distinct genetic entity. The most recent split (T_3) between G3 and G4 coincides with late MIS 3, a warm-humid interval that may have provided a final opportunity for *B. chinense* to cross northern latitudinal barriers and complete the last stage of stepwise range expansion. While these demographic estimates provide a useful temporal framework for interpreting the spatial genetic structure of *B. chinense*, they should be interpreted in light of the assumptions of SFS-based demographic inference, particularly that the analyzed loci are broadly neutral. Although Dataset 3 was designed to reduce the potential influence of selection by excluding candidate environment-associated loci and SNPs located within or near annotated genes, this filtering strategy remains indirect and may not completely remove all selected or linked loci. Accordingly, the fastsimcoal2-based results should be interpreted within the constraints of this filtering strategy.

Taken together, these results support a unified scenario in which Pleistocene climatic pulsing determined the timing of expansion and isolation, whereas complex topography controlled the spatial routing of dispersal and the persistence of lineage boundaries. Reconstruction of demographic histories indicates that lineages generally expanded during the mid-Pleistocene (~ 300 – 500 kya) and later stabilized, although the magnitude of expansion differed among groups. Notably, the long-term low effective population size and limited expansion signal in Group 1 suggest stronger or more prolonged bottlenecks, providing demographic support for repeated founder effects during stepwise range progression. In this framework, geographic barriers did not simply delimit lineages after divergence; rather, mountain chains, plateau margins, and fragmented terrain acted as long-term “physical filters” that regulated colonization, reduced subsequent gene flow, and preserved lineage independence even during climatically favorable intervals. Concordantly, chloroplast haplotypes show high within-region diversity but very limited sharing among regions, implying long-term regional persistence and restricted seed- or pollen-mediated connectivity across these barriers (Dellinger et al., 2022; Kartzin et al., 2013; Thia, 2014). These

results are consistent with a system in which climatic oscillations repeatedly created opportunities for movement, whereas topographic filtering and demographic founder effects translated that movement into sequential lineage turnover.

Overall, the stepwise genetic architecture of *B. chinense* is best interpreted as the outcome of Pleistocene climatic pulsing that repeatedly opened and closed dispersal opportunities, combined with persistent topographic filtering that partitioned dispersal and reduced gene flow. More generally, this pattern suggests that East Asian temperate floras were assembled not through simple continuous range tracking, but through repeated cycles of refugial persistence, staged expansion, and topographically filtered connectivity. In this context, the evolutionary history of *B. chinense* not only clarifies the phylogeography of a widespread medicinal herb, but also provides a temperate-herb perspective on a broader assembly model that links Quaternary climatic pulsing, regional dispersal corridors, and lineage diversification across East Asia.

Comparable phylogeographic patterns from other biodiversity hotspots suggest that the evolutionary trajectory of *B. chinense* is not an isolated East Asian case, but part of a broader biogeographic process in which Quaternary climatic pulsing, acting through topographically mediated isolation and episodic connectivity, generates lineage turnover and regional genetic structure (Kartzin et al., 2013; Thia, 2014). Within East Asia, this “stepwise expansion” model also has important implications for understanding the assembly of temperate floras. A comparable northward migration framework has been proposed for several East Asian temperate taxa, particularly deciduous broad-leaved forest species, in which southwestern or southern refugial persistence during glacial periods was followed by staged postglacial expansion and regional differentiation along major geomorphic and climatic corridors (Potts et al., 2013). In this sense, *B. chinense* extends a pattern more commonly discussed for woody plants to a widespread temperate herb, suggesting that stepwise expansion may represent a more general assembly process across East Asian temperate biota. At the same time, the demographic and phylogeographic signals observed here suggest that temperate herbs may not simply mirror woody-plant histories, because their responses to climatic pulsing and dispersal filtering can retain stronger founder effects and finer-scale regional turnover. Our results therefore help fill an important gap in evolutionary models for East Asian temperate herbs and further support a Pleistocene history of climatically pulsed expansion and topographically mediated isolation in *B. chinense*.

4.2. Isothermality-driven polygenic genomic adaptation

Climate-associated genomic signals were widespread in *B. chinense*, with isothermality (Bio3) showing the strongest associations among all environmental variables examined. Combining LFMM and BayPass analyses identified a large number of environment-associated SNPs, with broad concordance between methods, indicating that climatic factors account for a substantial proportion of genomic variation. Isothermality (Bio3), defined as the ratio of diurnal temperature range to annual temperature range, provides an integrative measure of short-term thermal stability versus variability (Gosling et al., 2014). *B. chinense* spans a broad climatic gradient from highland and inland regions with pronounced diurnal temperature fluctuations to comparatively thermally buffered lowlands in eastern China. Correspondingly, the spatial pattern of Bio3 closely matches the distribution of genomic signatures of selection (Manel et al., 2016). This concordance suggests that isothermality is not merely a descriptive climatic descriptor of habitat differentiation, but rather a core ecological selective pressure that has been persistently imposed during range expansion and establishment, thereby promoting local adaptive divergence (Parsons, 2005; Tattersall et al., 2012). Additional temperature and precipitation variables (Bio11, Bio15, Bio19) also exhibited widespread genomic signals.

Functional annotation of genes near significant SNPs, followed by

GO enrichment, revealed significant overrepresentation of processes related to stress responses, lipid metabolism, organelle and membrane organization, intracellular localization and transport, DNA repair, and cytoskeletal assembly and regulation (Fig. 4; Table S9). From a physiological perspective, environments characterized by large diurnal temperature ranges and frequent short-term thermal fluctuations impose persistent challenges to membrane stability, metabolic rates, photosynthetic performance, and DNA integrity, thereby generating strong natural selection (Parsons, 2005; Tattersall et al., 2012). Accordingly, the enriched functional modules encompassing “membrane and lipid metabolism–stress defence–repair and maintenance–transmembrane and intracellular transport” likely act in concert to enable rapid adjustment and homeostatic regulation in response to alternating heat–cold conditions and water-availability fluctuations over short time scales. This pattern is consistent with a common strategy in temperate plants, whereby environmental adaptation is achieved primarily through coordinated fine-tuning across complex regulatory networks rather than through single, large-effect mutations (Marand et al., 2023).

Mapping Bio3-significant SNPs to gene bodies yielded 533 candidate genes. Functional annotation showed enrichment for transmembrane transport, core metabolism, transcriptional and chromatin regulation, lipid metabolism, vesicle trafficking, and stress and defense; representative genes include HDH, TAF2, DGK7, ABCC2, BIG5, TPS1, and RRP44A (Table S10). This module composition is consistent with the eggNOG-mapper-based enrichment results (Fig. 4; Table S9), indicating a multilevel, coordinated pattern of environmental adaptation (An and Ning, 2025; Hanssen et al., 2013; Verkerk et al., 2015). Candidate sets linked to different climatic variables exhibited distinct functional emphases: Bio3 (isothermality) favored genes coupling metabolism with epigenetic regulation, suggesting that diurnal and seasonal temperature regimes drive adaptation through coordinated adjustments in carbon and lipid metabolism together with chromatin remodeling; Bio15 (precipitation seasonality) highlighted transport and vesicle systems and kinase cascades, consistent with rapid solute allocation and the timely deployment of receptors and transporters under fluctuating water availability; Bio19 (precipitation of the coldest quarter) emphasized membrane homeostasis and defense and repair, accompanied by stabilization at the RNA and transcriptional levels. Importantly, these results also suggest that adaptation in *B. chinense* is unlikely to rely on genomic hard-wiring or phenotypic plasticity alone, but rather on their complementarity. On the one hand, polygenic changes in metabolism, transmembrane transport, and transcriptional or chromatin regulation provide a relatively stable genomic framework for coping with recurrent thermal variability. On the other hand, downstream trait adjustment, including variation in growth-related traits such as plant height, likely allows populations to respond flexibly to short-term environmental heterogeneity. Together, a genetically embedded adaptive architecture and plastic trait expression may enhance the evolutionary resilience of widespread species by enabling both long-term persistence and short-term responsiveness across heterogeneous environments. (Perkowski, 2012; Wilton, 2012).

4.3. Environmental plasticity, geographic structure, and adaptive trade-offs of plant height

The present-day genetic structure of *B. chinense* appears to have been shaped primarily by geographic isolation, with environmental filtering providing an additional but weaker contribution. Within this structured genomic background, K-means clustering divided populations into primarily eastern (Group A) and western (Group B) clusters that differ not only in Bio3 but also in plant height, with Group A exhibiting greater mean height than Group B. Among the Bio3-associated candidates, we identified 30 loci related to plant height. In the proximal growth modules, five genes involved in auxin polar transport (ABCB15, PHOT1, ABCB11, PILS6, ABCB25) and three genes involved in cell-wall extension (CESA8, XTH7, CESA2) were highlighted (Fig. 4). Chloroplast and

height-candidate haplotypes (e.g., XTH7, MOR1) show partial geographic structuring that broadly corresponds to the east–west grouping and plant-height differences, suggesting that climatic factors, particularly isothermality (Bio3), may have contributed to morphological differentiation on top of an already structured geographic background (Li et al., 2023).

Spatially, individuals from low-Bio3 regions tend to be taller; however, the haplotype networks do not show complete group-specific fixation, and the observed pattern is therefore more consistent with a combination of environmentally responsive variation and partial genetic structuring than with simple fixed genetic divergence alone. Although 30 nuclear genes related to height (spanning transcription and growth regulation, auxin synthesis/transport/signaling, sugar/energy metabolism, and cell-wall extension) were recovered among Bio3-associated candidates, these loci are better interpreted as candidate components of a polygenic, environmentally responsive system rather than as fixed markers that fully distinguish the two groups. Such a pattern is consistent with an ecological resource-allocation trade-off: under low Bio3 (smaller diurnal thermal range and a more continuous thermal window), prolonged periods of carbon assimilation and turgor favor taller growth to enhance light interception and reproductive opportunity; under high Bio3, the costs of maintaining membrane and protein homeostasis and stress metabolism increase, shifting resources from elongation to stability and defense, thereby constraining height (Poiré, 2011; Singhal et al., 2024). This growth–robustness trade-off explains the spatial gradient in plant height alongside incomplete genetic partitioning, and underscores the complementarity between genomic adaptation and phenotypic plasticity. Environmental PCA further shows clear climatic differentiation among regions, while the Mantel results indicate that geographic distance was more strongly associated with genetic divergence than environmental distance, although both associations were significant. We therefore infer that long-term geographic barriers likely contributed substantially to population structure, whereas climatic gradients, particularly temperature stability and precipitation seasonality, may have further contributed to local divergence and associated variation in plant height. (Ohsawa and Ide, 2008). In this context, climatic drivers such as Bio3 influence the environment–trait–adaptation relationship by modulating the balance between growth and robustness.

Overall, Bio3 appears to operate at the genomic level through a polygenic network involving metabolism, transmembrane transport, and stress responses, whereas variation in plant height likely reflects both environmentally responsive phenotypic adjustment and an underlying contribution from partial genetic structuring. Importantly, the cpDNA network is interpreted here as reflecting demographic and phylogeographic structure, whereas candidate nuclear loci such as XTH7 and MOR1 are considered putative contributors to height variation. Under varying temperature regimes and hydrothermal conditions, plant height is therefore likely shaped by both plastic responses and polygenic genomic effects mediated through growth-control pathways such as auxin-polarity-driven transport and cell wall and cytoskeletal extension, rather than by simple stable genetic fixation across populations. Overall, the present-day genetic structure of *B. chinense* is best understood as the joint outcome of geographic isolation and environmental filtering, whereas plant-height variation appears to reflect environmentally responsive variation superimposed on this structured genomic background, with plastic and genetic components both likely contributing.

5. Conclusions

By integrating population genomics and environmental association analyses, this study reveals that *B. chinense* experienced stepwise divergence from Southwest China toward North China and Northeast China during the Pleistocene, accompanied by asymmetric gene flow and climatically pulsed range expansion followed by topographic isolation. Partial Mantel tests showed that both geographic and

environmental distances were significantly associated with genetic structure, although the geographic association was stronger, suggesting that topographic barriers and environmental heterogeneity jointly shaped regional divergence. At the adaptive level, isothermality (Bio3) was strongly associated with genomic variation, and candidate genes were enriched in functions related to membrane and lipid metabolism, transmembrane transport, stress defense and repair, and transcriptional or chromatin regulation, consistent with polygenic adaptation to climatic instability. Plant height varied spatially along Bio3 and hydrothermal gradients, and appears to reflect environmentally responsive variation together with an underlying contribution from partial genetic differentiation. Overall, our results provide a genomic perspective on how widespread temperate plants persist, diverge, and adapt across complex climatic and topographic gradients in East Asia.

CRedit authorship contribution statement

Yong-Xiu Song: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Yi Shi:** Visualization, Software. **Jia-Qing Hou:** Visualization, Software. **Zhen Zhang:** Visualization, Methodology. **Xin-Yi Yu:** Visualization, Methodology. **Ying-Qi Zeng:** Investigation. **Xiao-Yu Jiang:** Investigation. **Yan Yu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.palaeo.2026.113875>.

Data availability

The authors confirm that all data necessary for supporting the scientific findings of this paper have been provided.

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